Example Diagnostic Measures Logistic Regression

Singh

##Read the Example3 data

```
library(readxl)
uisdat <- read_excel("C:/Users/Pradeep Singh/OneDrive - Southeast Missouri State University/Teaching/Spring2024/CSDA5110/Week6/logistic regression/UMARU.xlsx")
View(uisdat)
head(uisdat)
```

```
## # A tibble: 6 × 9
##      ID   Age  BECK  IVHX NDRUGTX  Race Treat  Site DFREE
##   <dbl> <dbl> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     1    39   9       3       1     0     1     0     0
## 2     2    33  34       2       8     0     1     0     0
## 3     3    33  10       3       3     0     1     0     0
## 4     4    32  20       3       1     0     0     0     0
## 5     5    24   5       1       5     1     1     0     1
## 6     6    30  32.6     3       1     0     1     0     0
```

Residuals for Logistic Regression

```
library(api2lm)
library(aod)
library(MASS)
intercept_Model <- glm(DFREE~ 1, family=binomial, uisdat)
full_model <- glm(DFREE~ Age+BECK+IVHX+NDRUGTX+Race+Treat+Site, family=binomial, uisdat)
summary(full_model)
```

```
##
## Call:
## glm(formula = DFREE ~ Age + BECK + IVHX + NDRUGTX + Race + Treat +
##     Site, family = binomial, data = uisdat)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.9991800  0.5879337  -3.400 0.000673 ***
## Age          0.0482398  0.0172169   2.802 0.005080 **
## BECK         0.0004193  0.0107988   0.039 0.969030
## IVHX        -0.3699491  0.1283375  -2.883 0.003944 **
## NDRUGTX     -0.0617824  0.0257134  -2.403 0.016273 *
## Race         0.2296271  0.2229677   1.030 0.303072
## Treat        0.4292549  0.1985299   2.162 0.030605 *
## Site         0.1212729  0.2152053   0.564 0.573080
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 653.73  on 574  degrees of freedom
## Residual deviance: 620.09  on 567  degrees of freedom
## AIC: 636.09
##
## Number of Fisher Scoring iterations: 4
```

```
res_pearson <- residuals(full_model, type ="pearson")
res_deviance <- residuals(full_model, type ="deviance")
head(res_pearson, 10)
```

```
##          1          2          3          4          5          6
7
## -0.6516713 -0.5493843 -0.5301968 -0.4451359  1.5367256 -0.5271042
4.2440901
##          8          9         10
## -0.4731536 -0.6302210 -0.5053179
```

```
head(res_deviance, 10)
```

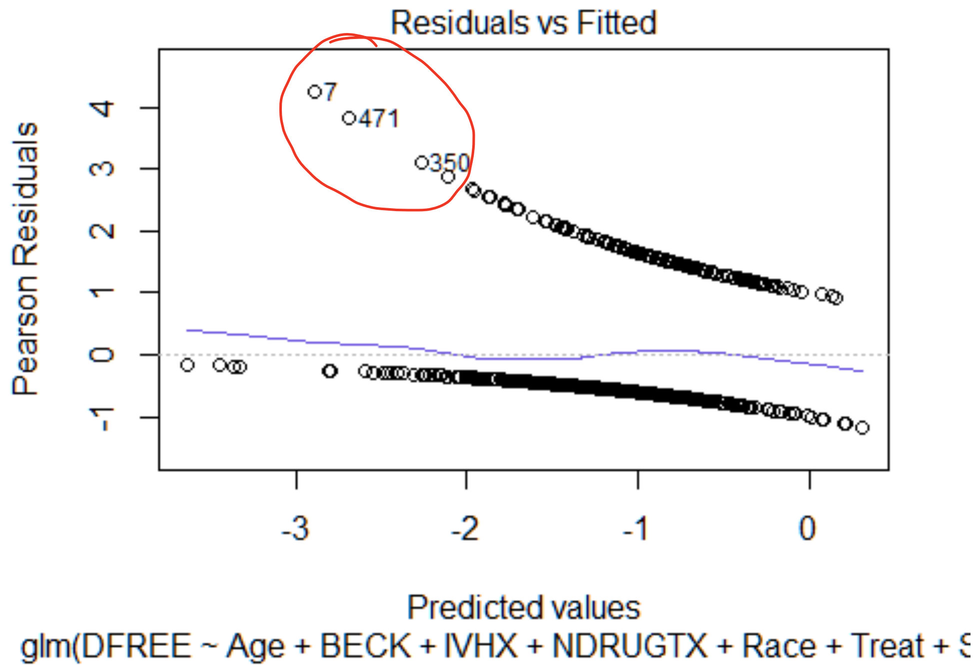
```
##          1          2          3          4          5          6
7
## -0.8413609 -0.7263135 -0.7038833 -0.6012908  1.5571737 -0.7002431
2.4269677
##          8          9         10
## -0.6356438 -0.8178690 -0.6744055
```

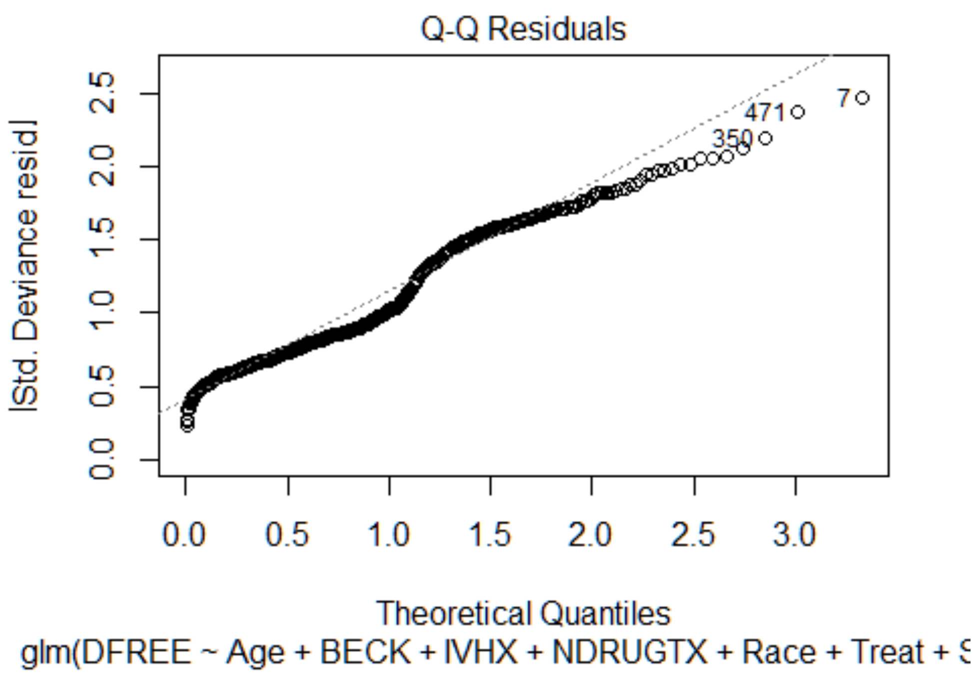
##studentized residuals, Hat, Dfbeta, dffit, cooks D Plots

```
library(api2lm)
library(aod)
library(MASS)
library(car)
```

```
## Loading required package: carData
```

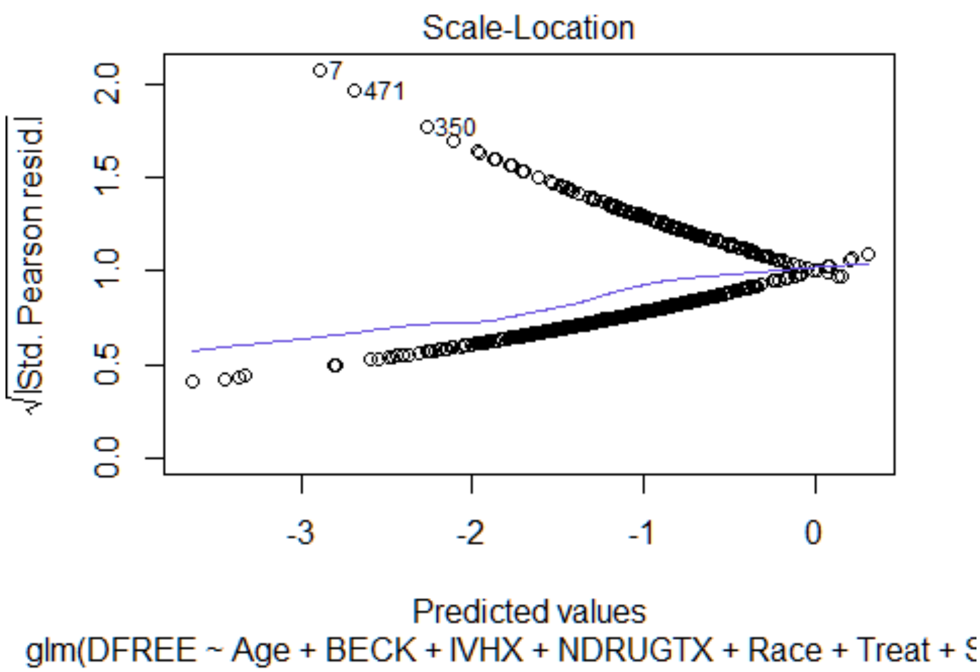
```
intercept_Model <- glm(DFREE~ 1, family=binomial, uisdat)
full_model <- glm(DFREE~ Age+BECK+IVHX+NDRUGTX+Race+Treat+Site, family=
binomial, uisdat)

plot(full_model)
```

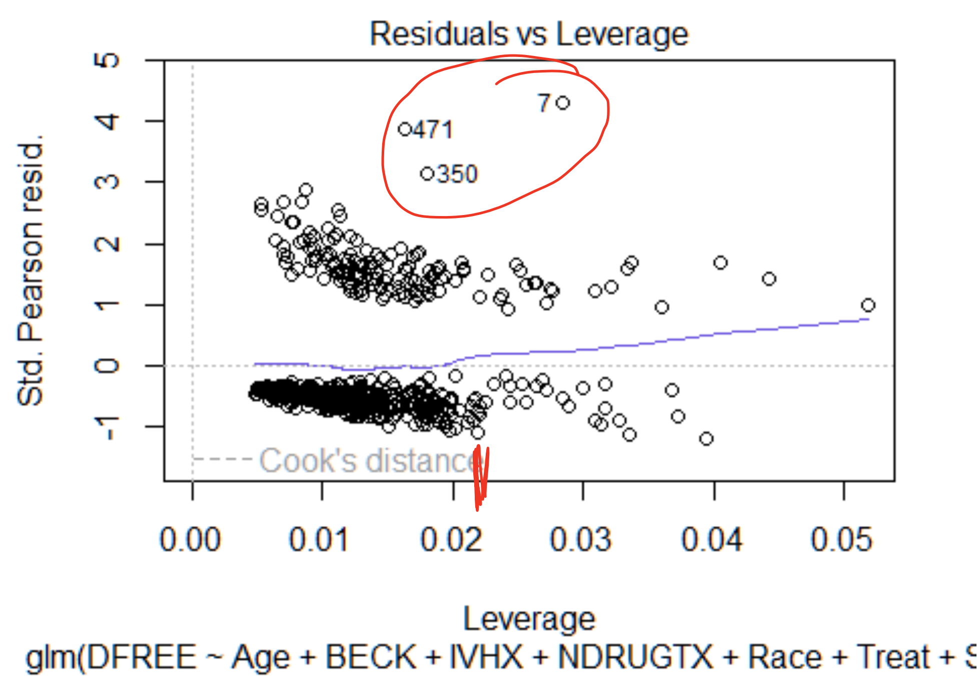




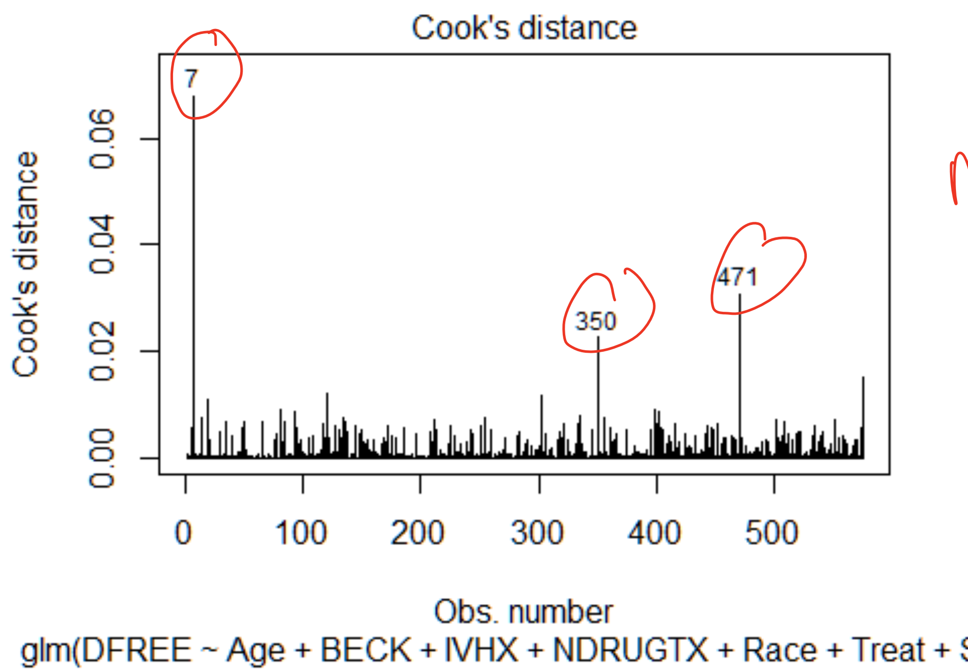
$h_{ii} = \frac{2(p+1)}{n}$

$\frac{2 \times 7}{575} = .024$






```
plot(full_model, which=4)
```



none of these are greater than .5

```
marginalModelPlots(full_model)

## Error in smooth.construct.tp.smooth.spec(object, dk$data, dk$knots) :
##   A term has fewer unique covariate combinations than specified maximum degrees of freedom
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```
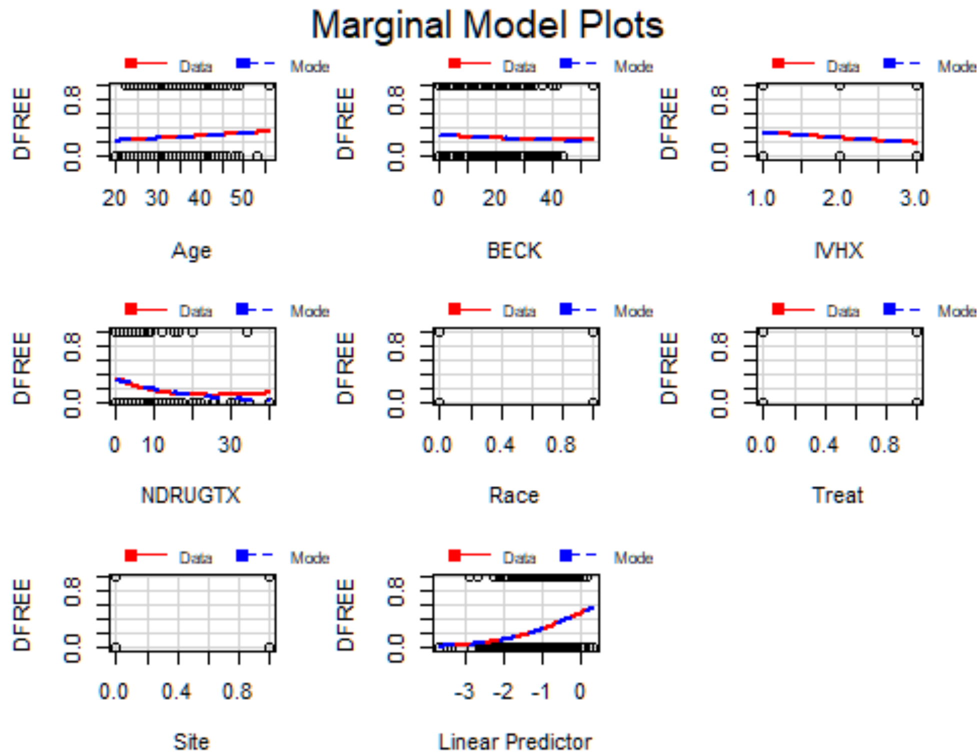



Blue line should fall on red line

##studentized residuals, Hat, Dfbeta, dffit, cooks D

```
library(api2lm)
library(aod)
library(MASS)
library(car)

intercept_Model <- glm(DFREE~ 1, family=binomial, uisdat)
full_model <- glm(DFREE~ Age+BECK+IVHX+NDRUGTX+Race+Treat+Site, family=
binomial, uisdat)

inf_dfb <- dfbeta(full_model)
inf_dffit <- dffits(full_model)
inf_cook <- cooks.distance(full_model)
inf_hat <- hatvalues(full_model)
inf_stud <- studres(full_model)
inf_stats <- cbind(inf_dfb, inf_dffit, inf_cook, inf_hat, inf_stud)
head(inf_stats, 10)

##      (Intercept)           Age          BECK         IVHX       NDRUGTX
## 1    0.012245713 -5.071791e-04  4.147026e-04 -0.005476820  0.0012041398
## 2    0.008794201 -1.043346e-04 -5.830570e-04  0.002399488 -0.0007471067
## 3   -0.005218065  1.716662e-04  3.170047e-04 -0.005002648  0.0004111411
## 4   -0.006311011  1.649561e-04 -2.255632e-05 -0.003845088  0.0005971958
## 5    0.071091366 -1.694974e-03 -1.087145e-03 -0.002558407  0.0015347044
## 6    0.002919176  2.668503e-04 -4.417126e-04 -0.004740419  0.0007959685
## 7   -0.013038603 -3.063531e-04 -9.299729e-05 -0.006125645  0.0104031368
```

```
## 8   -0.016483764   5.716273e-04   2.948119e-04  -0.005540529   0.0004195154
## 9    0.030916140  -6.953330e-04  -4.760570e-04  -0.002944329   0.0007432044
## 10   0.010118376  -9.537598e-05  -1.904086e-04  -0.002444681  -0.0002563595
##                Race         Treat          Site    inf_dffit      inf_cook
inf_hat
## 1    0.005365294  -0.007724270   0.0062974203  -0.09770021  0.0007833040
0.014335773
## 2    0.005292391  -0.005480877   0.0035713102  -0.07739465  0.0004689234
0.012129408
## 3    0.003241732  -0.005459357   0.0040076890  -0.06476097  0.0003256112
0.009098630
## 4    0.002023693   0.004725245   0.0030872108  -0.05053455  0.0001915524
0.007616432
## 5    0.022497682   0.010321275  -0.0096479004   0.20444763  0.0055526421
0.018134315
## 6    0.003305010  -0.005234657   0.0030692914  -0.07688136  0.0004582889
0.012858642
## 7   -0.004238716   0.009326806   0.0008752881   0.40426825  0.0677740857
0.028414901
## 8    0.002203099  -0.004382334   0.0031702217  -0.06438581  0.0003143554
0.010987778
## 9    0.005483049  -0.007280161   0.0046080132  -0.09165395  0.0006823317
0.013378306
## 10   0.003309860  -0.005058588   0.0027523411  -0.05534499  0.0002353276
0.007266065
##        inf_stud
## 1   -0.6492439
## 2   -0.5466818
## 3   -0.5267610
## 4   -0.4418846
## 5    1.5366108
## 6   -0.5246796
## 7    4.3279673
## 8   -0.4705119
## 9   -0.6275564
## 10  -0.5015759
```

```
influence_stats(full_model)
```

```
##            cooks    leverage  studentized
## 7    0.06777409  0.02841490     2.533172
## 471  0.03083978  0.01625118     2.397713
## 350  0.02254548  0.01799288     2.216161
## 575  0.01530698  0.04051463     1.666822
## 120  0.01224090  0.03361930     1.649917
## 303  0.01170248  0.04418104     1.497836
```

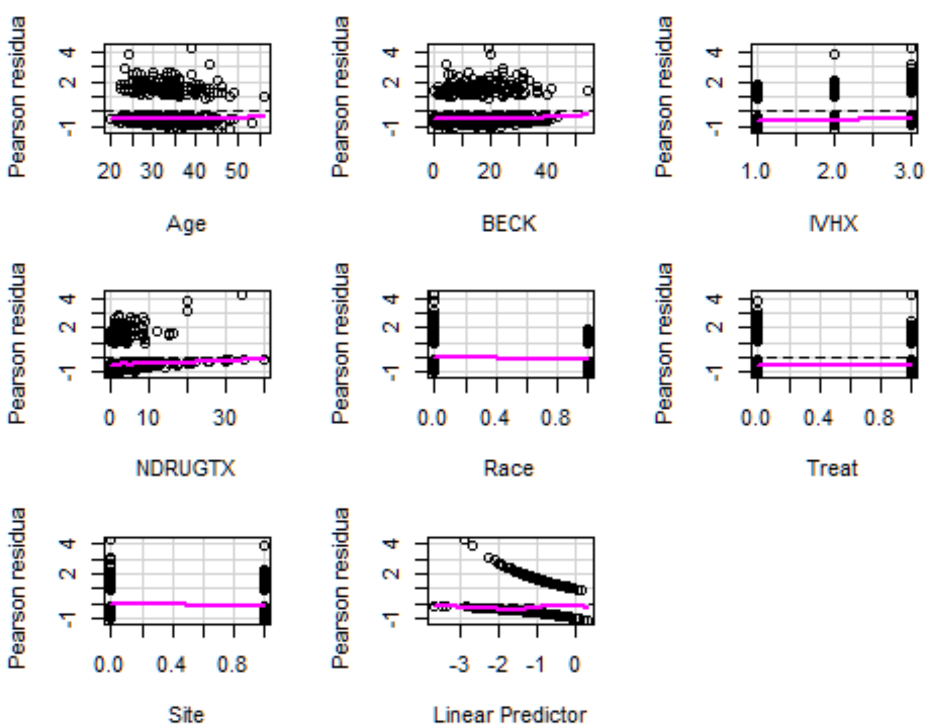
Influence Tests

```
library(api2lm)
library(aod)
```

```r
library(MASS)
library(car)
library(DescTools)
```

```
##
## Attaching package: 'DescTools'

## The following object is masked from 'package:car':
##
##     Recode
```

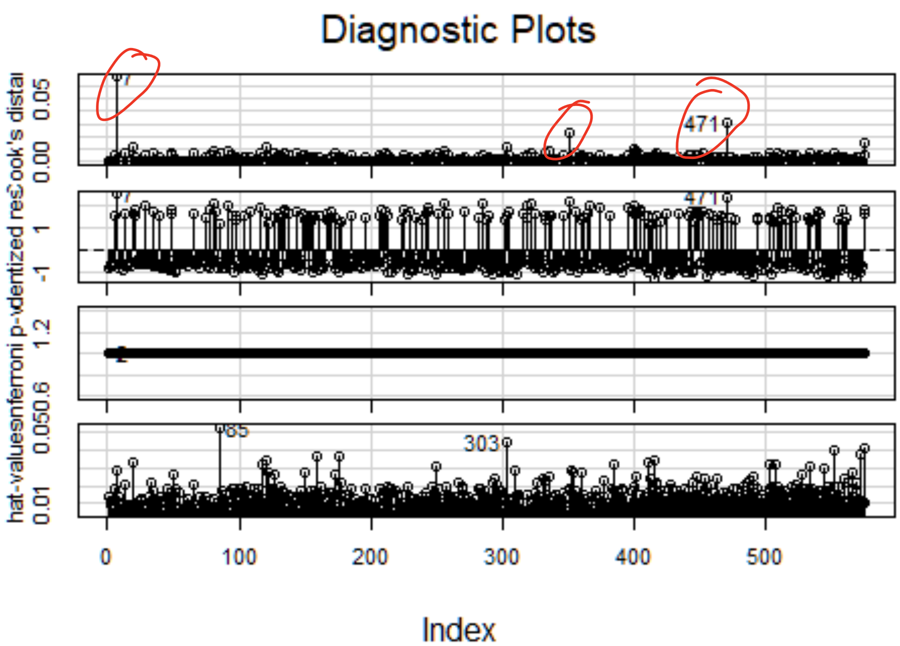
```r
intercept_Model <- glm(DFREE~ 1, family=binomial, uisdat)
full_model <- glm(DFREE~ Age+BECK+IVHX+NDRUGTX+Race+Treat+Site, family=
binomial, uisdat)

residualPlots(full_model)
```




```
##           Test stat Pr(>|Test stat|)
## Age          0.3122           0.5763
## BECK         0.9449           0.3310
## IVHX         0.8449           0.3580
## NDRUGTX      2.0158           0.1557
## Race         0.0000           1.0000
## Treat        0.0000           1.0000
## Site         0.0000           1.0000
```

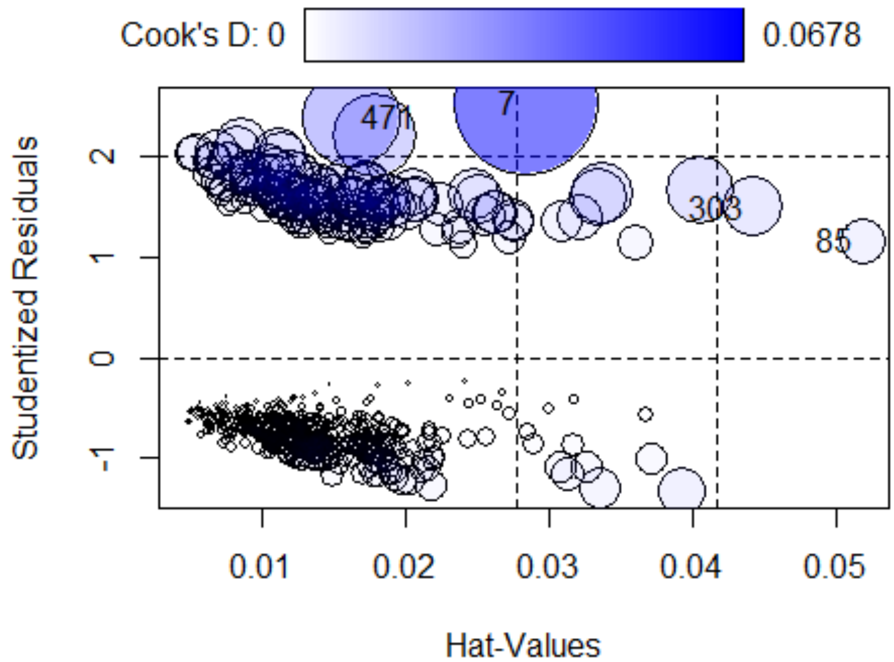
none of them are significantly affected by outliers

```r
influenceIndexPlot(full_model)
```




```
outlierTest(full_model)

## No Studentized residuals with Bonferroni p < 0.05
## Largest |rstudent|:
##    rstudent unadjusted p-value Bonferroni p
## 7  2.533172           0.011304           NA

influencePlot(full_model)
```

outlier
but not influential




```
##          StudRes        Hat       CookD
## 7      2.533172 0.02841490 0.067774086
## 85     1.170973 0.05176384 0.006727927
## 303    1.497836 0.04418104 0.011702477
## 471    2.397713 0.01625118 0.030839781
```

```
PseudoR2(full_model)
```

```
##   McFadden
## 0.05145319
```

```
VIF(full_model)
```

```
##       Age     BECK     IVHX  NDRUGTX     Race    Treat     Site
## 1.206692 1.037101 1.334433 1.132680 1.064418 1.007101 1.056226
```

greater than 5
implies Multicollinearity